

Addressing the critical limitations of the standard neuron model in artificial neural networks through a bio-inspired update

Raul Mohedano^{1@}, Thomas Batard², Erik Velasco-Salido¹, Ramsses De Los Santos Mendoza², Jorge H. Martinez², Stacey Levine³, Marcelo Bertalmío¹
@: raul.mohedano@csic.es 1: Spanish National Research Council (CSIC), Spain; 2: Center for Research in Mathematics (CIMAT), Mexico; 3: National Science Foundation, USA

Background and motivation

Power of current AI: **artificial neural networks (ANN)**

👍 Outstanding performance and applications in multiple modalities

👎 Poor alignment to brain functionality [3]

👎 They exhibit certain **critical limitations** which are

- Not solvable by upscaling
- **Not present in living beings**

High sensitivity to input perturbations

Adversarial attacks [4]:

- Intentional, malicious perturbations
- Imperceptible to humans
- Unacceptable in high-risk tasks

Need for vast amounts of training data

Fields with inherent data scarcity, e.g.:

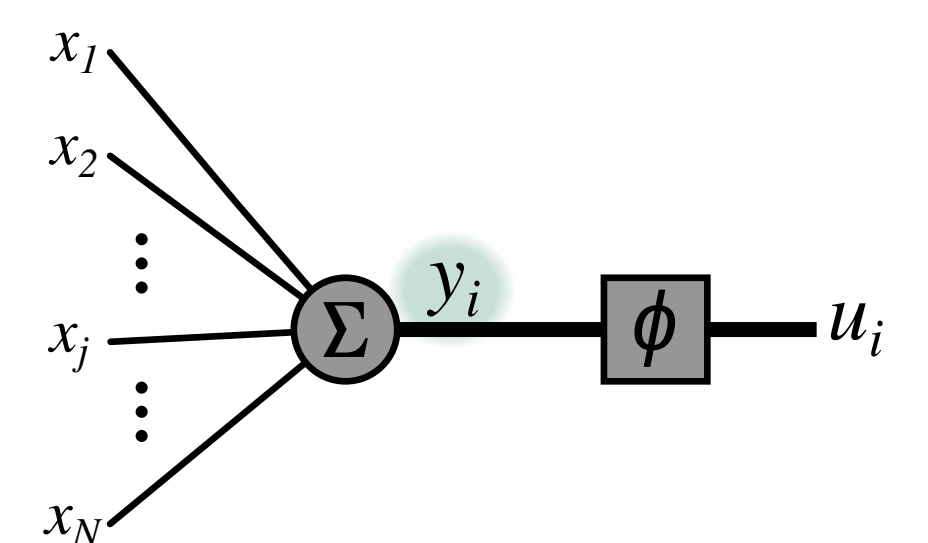
- Due to lack of experts/ground truth
- Due to insufficient subjects/population

ANN units/blocks correspond to the **point-neuron model**:

- **Origin, late 1950's: inspiration in neuro** (visual system [1])
- **Still today, unchanged:**
 - virtually **all modalities** and applications
 - **standard model (SM)** of deep learning

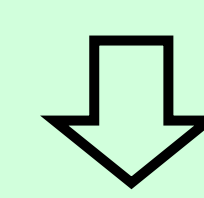
$$y_i = \sum_{j=1}^N m_{ij} x_j - b_i$$

$$u_i = \phi(y_i)$$



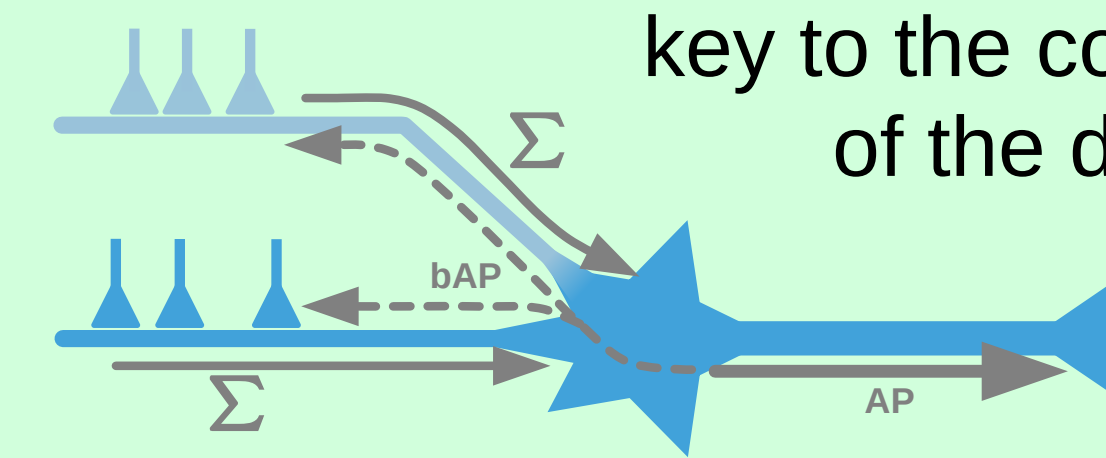
Hypothesis

The **oversimplifications** of the **point-neuron model** are the cause !!!



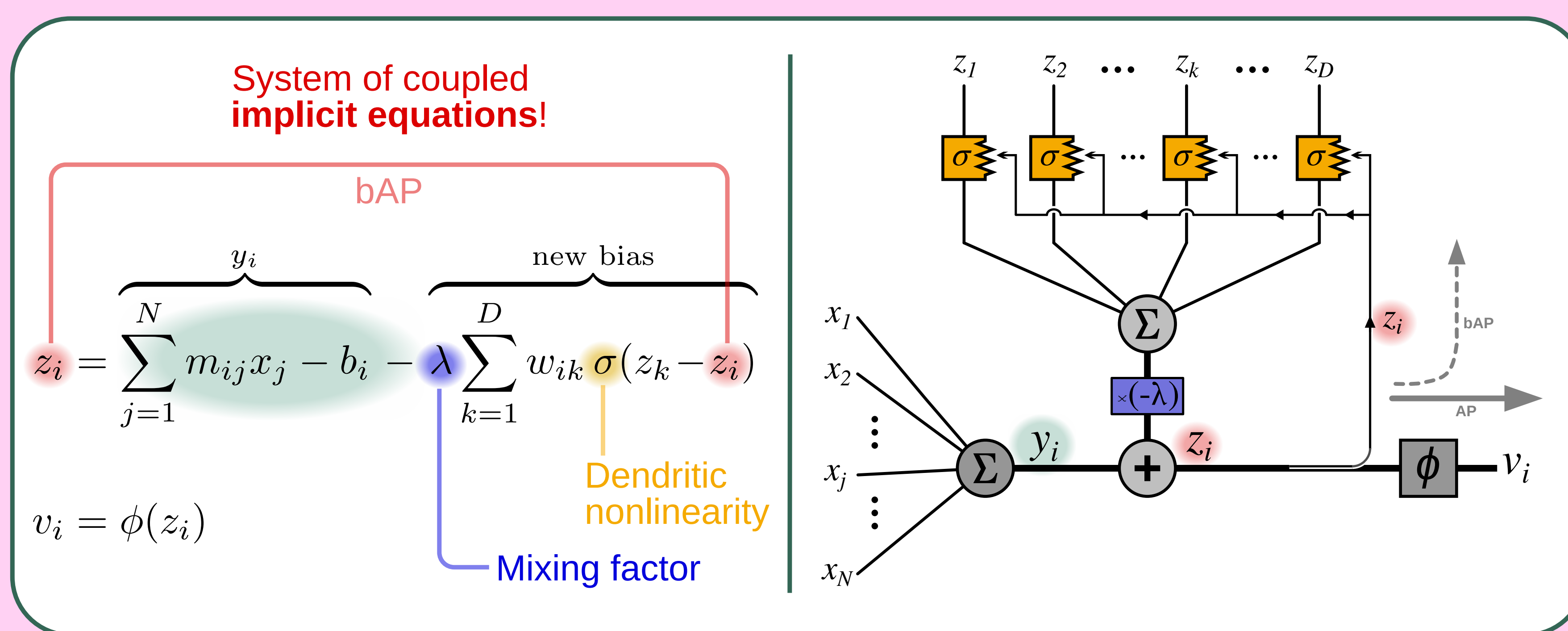
Biological inspiration: bAP

- **Backpropagating action potentials (bAP)** key to the computations of the dendrites [5]



- Most bio-inspired models still rely on the SM and are unsuitable for representation of the bAP
- bAP studied and evaluated with success in a recent model [2] for V1 neurons
- Here: the **first neuron model considering the bAP**
 - Calculations: through an **implicit formulation**

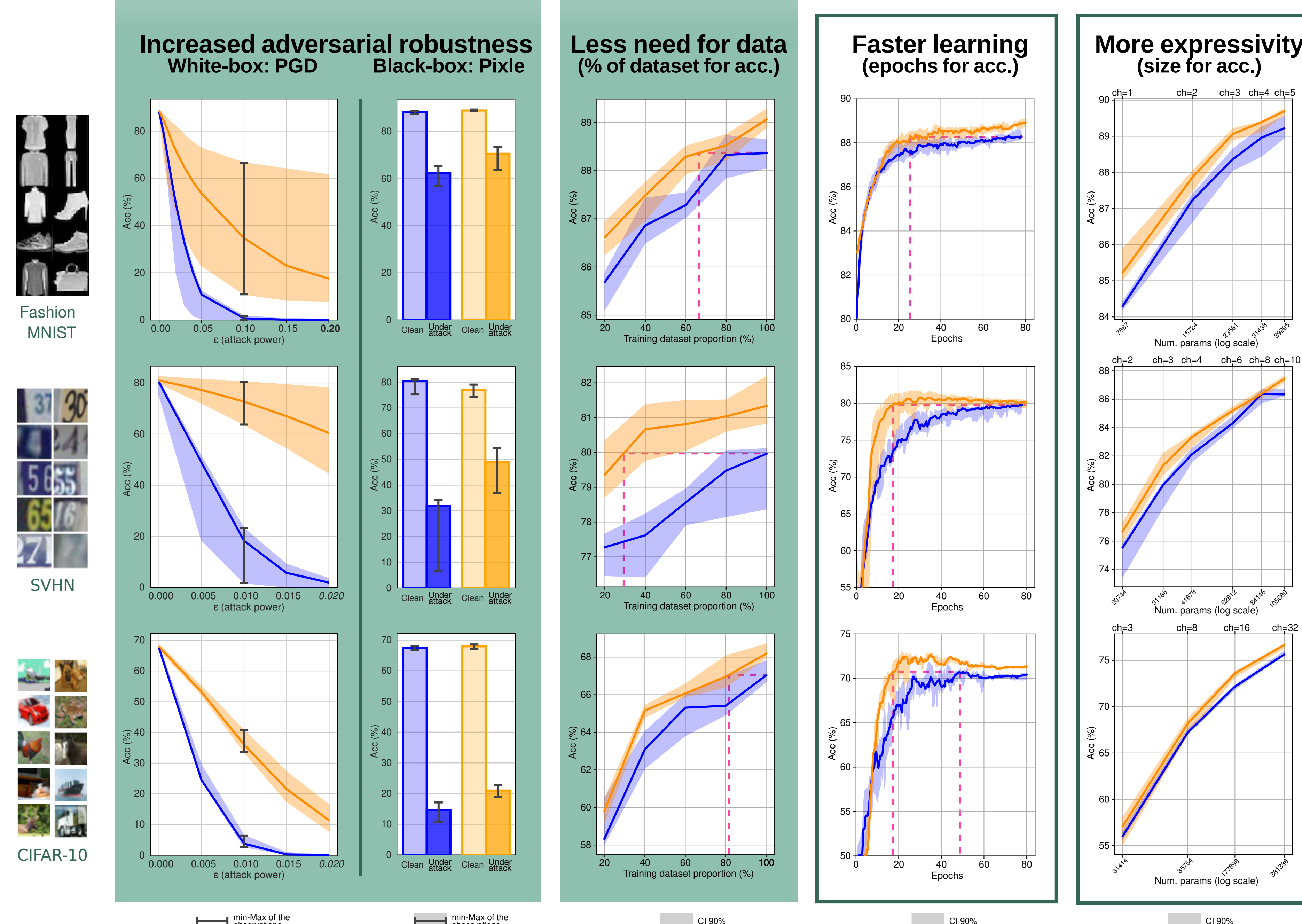
Proposal: Implicit Bias Neural Networks (IBNN)



Experimental results

Each experiment: — Standard model (SM) — Proposed IBNN model

- Identical architecture and number of parameters for SM/IBNN
- Only difference: SM vs IBNN in the hidden layers



Discussion

- **Inspiration from neuroscience** shows potential for **overcoming the critical limitations** of the **standard deep learning** model
- bAP-like effects can be incorporated in artificial neurons via implicit formulations
- The **proposed IBNN** neuron achieves
 - **clear improvement in the critical limitations:**
 - Better **robustness to adversarial attacks**
 - Similar **accuracy** for **reduced training data**
 - **significant improvement** in other aspects, e.g.:
 - Better **accuracy** after **less epochs**
 - Better **accuracy** using **smaller networks**
- Other existing bio-inspired models have considered, like the proposal, nonlinear dendrites and implicit/PDE-based layers: the **difference in the results** should be attributed to the differential aspect IBNN, i.e. the **inclusion of the bAP**
- Although the proposed neuron model and the conducted experiments correspond to vision and are restricted to image data, a similarly positive behavior in other modalities is expected by analogy to the origin of the standard model [1]
- Positive effects depend on the mixing factor λ : need for mechanisms for optimally establishing λ for multiple aspects of interest

References

- [1] Hubel & Wiesel, Receptive fields of single neurones in the cat's striate cortex. *J. Physiol.* 148, 574–591 (1959)
[2] Rentzeperis, Prandi & Bertalmío, A neural model for V1 that incorporates dendritic nonlinearities and back-propagating action potentials. *J. Neurosci.* (2025)

[3] Bowers et al., Deep problems with neural network models of human vision. *Behav. Brain Sci.* 1–74 (2022)

[4] Wichmann & Geirhos, Are deep neural networks adequate behavioral models of human visual perception? *Annu. Rev. Vis. Sci.* 9, 501–524 (2023)

[5] Larkum, Are dendrites conceptually useful? *Neurosci.* 489, 4–14 (2022)

